Hemanth Rachaboyina

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SKILLS

Languages: Java, Python, SQL, Golang

Cloud Technologies: Amazon Web Services (AWS), GCP

AI Technologies: Machine Learning, Deep Learning, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Transformer Models, Fine-Tuning LLMs (LoRA, QLoRA), Vector Databases (FAISS, ChromaDB)

Tools & Frameworks: FastAPI, LangChain, LangGraph, LlamaIndex, CrewAI, TensorFlow, PyTorch, OpenCV, Power BI

WORK EXPERIENCE

Kautilya AI (startup), Bangalore Aug 2024 – Oct 2025

AI Engineer

- Evaluated and compared multiple agentic AI frameworks — LangChain, LangGraph, AutoCrew, and LlamaIndex through hands-on prototyping, and selected LlamaIndex as the core framework based on its superior retrieval integration and tool-calling flexibility for financial data workflows.
- Designed and built a multi-agent AI system for a financial trading copilot, architecting 8+ specialized agents handling market data parsing, decision reasoning, and response generation within a 5-person engineering team.
- Built a production RAG pipeline from scratch using vector embeddings, semantic chunking, and hybrid search (dense + sparse retrieval) to ground LLM responses in knowledge base significantly improving response relevance over baseline.
- Implemented Model Context Protocol (MCP) to manage state and communication across distributed agents, reducing context loss between multi-step reasoning tasks.
- Owned prompt engineering and iterative testing cycles designing system prompts, evaluating outputs, and refining agent behavior based on failure case analysis.

Tech Stack: *Python, RAG, Fine-tuning, FastAPI, Agentic AI, MongoDB, Prompt Engineering*

EDUCATION

Blekinge Institute of Technology, Sweden — Master's in Software Engineering
Grade: Very Good

JNTUH College of Engineering, Hyderabad — Bachelor's in Computer Science
GPA: 8.3

PROJECTS

SpiritPrep – AI Interview Simulator

https://spirit-prep-ai.vercel.app/

- Built a full-stack interview preparation platform where users practice mock interviews based on a chosen topic or by pasting a job description — the system automatically generates a relevant, adaptive question set using GPT.
- Fine-tuned a GPT model on OpenAI's platform using a custom dataset of 5,000 AI-generated interview Q&A examples, improving response quality and domain relevance over vanilla GPT prompting.
- Integrated LemonFox API for both STT and TTS — users speak their answers naturally and hear AI responses back, creating a realistic voice-based interview experience.
- Built a FastAPI backend to handle session management, question sequencing, and AI feedback generation, connected to a React frontend for a clean interview UI.

- Implemented a post-interview feedback engine that evaluates user responses and returns personalized performance insights and improvement suggestions.
- Currently used by 70+ real users running mock interviews locally.

VentureOS — The AI Co-Founder Agent

https://agent-cofounder-frontend-9tip.vercel.app/

- Architected a multi-agent AI system using LangGraph that autonomously conducts end-to-end startup validation — covering market research, SWOT analysis, pitch deck generation, and execution planning from a single idea input.
- Integrated real-time web search into the agentic pipeline to ground research in current market data, enabling dynamic, domain-aware insights rather than static LLM knowledge.
- Applied advanced prompt engineering techniques across specialized agent nodes to precisely steer LLM behavior for each analytical task, ensuring structured, role-specific outputs at every stage of the workflow.

GraphRAG Cognitive Pipeline

- Built and deployed a high-performance GraphRAG backend using FastAPI, Neo4j, and GPT-4o, transforming unstructured PDFs into structured knowledge graphs for advanced AI reasoning.
- Engineered an autonomous ingestion pipeline capable of entity extraction, relationship mapping, and ontology generation, enabling accurate multi-hop contextual retrieval beyond traditional vector search systems.
- Combined OpenAI semantic embeddings with optimized Cypher graph traversals to deliver highly precise, low-hallucination responses through deterministic knowledge synthesis and relational reasoning.

AWARDS AND CERTIFICATES

- Data Science Course on Coursera
- Solved Neetcode 150 Problems